Cycling of Ethanol and Acetaldehyde in Estuarine Waters

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Introduction

Increased use of ethanol as a fossil fuel substitute and additive is expected to have an impact on ethanol and acetaldehyde levels in the atmosphere and changes in the tropospheric concentrations of these species is likely to have an impact on oxidative capacity. To understand future impacts, we need to understand current tropospheric budgets. Current tropospheric budget estimates for both species have significant uncertainties; one of the largest sources of uncertainty is the role of the oceans and surface waters in cycling these species into and out of the troposphere. Our understanding is limited by our understanding of the processes that control concentrations in seawater and surface waters. Aqueous phase concentrations of ethanol and acetaldehyde are controlled by a complex interplay between biological sources and sinks, photochemical sources and sinks and air-water exchange. For small, oxygenated hydrocarbons the dominant source is generally photochemical and the dominant sink is generally biological uptake. Ethanol can also be converted to acetaldehyde biologically in the dark. Ethanol and acetaldehyde biological degradation rates and the rate of biological production of acetaldehyde from ethanol were measured at multiple sites over 18 months in a classic Southern California estuary. To aid interpretation a range of ancillary water quality measurements were also made. The microbial community was characterized by 16s rDNA sequencing and fluorescence microscopy and correlations between biological rates and community structure were explored.

Methods

- Water samples were collected from surface waters at the shoreline. One site typically sampled per week; sites rotated.
- Water quality parameters were measured in situ using a handheld probe.
- For bacteria counting, 20 mL water sample was filtered through black 0.2 µm polycarbonate filters, fixed with aldehydes, stored at 4oC then DAPI stained and counted using fluorescence microscopy.
- Microorganisms were extracted on 0.2 µm filters (200 ml) and the microbial community characterize by amplicon sequencing of 16S rDNA.
- For DOC analyses, water samples were 0.2 um filtered, acidified, refrigerated and analyzed by Shimadzu TOC analyzer.
- For optical properties, fluorescence and absorbance spectra are measured on 0.2 um filtered water samples using an Aqualog spectrometer (Horiba).
- For ethanol and acetaldehyde rate measurements, coarse filtered (40 µm) water samples were spiked with isotopically labelled compounds and incubated in the dark in head space less glass syringes. Periodic aliquots were then 0.2 um filtered and analyzed by purge and trap GC-MS. Rates were obtained from changes in concentration as a function of time.

Study Site: Newport Back Bay, Southern California

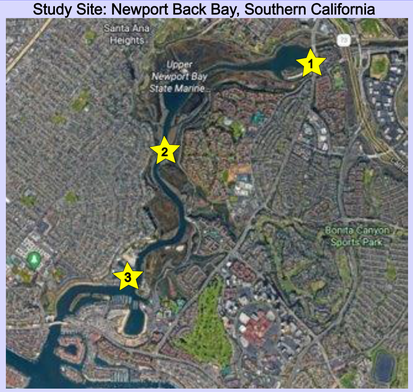


Figure 1. Map of the Newport Back Bay estuary. Site 1 is at the inflow and site 3 at the mouth, opening into the Pacific.

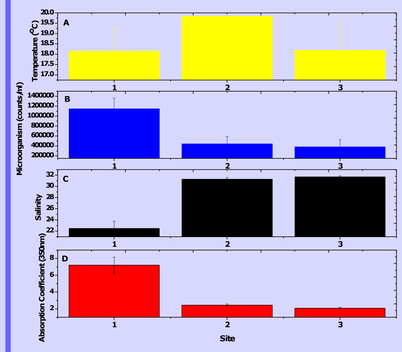


Figure 2. Average water quality parameters and bacteria counts by site. Site 1, at the river inflow to the estuary, has higher bacteria counts, lower salinity and higher absorbance. DOC was also higher at site 1.

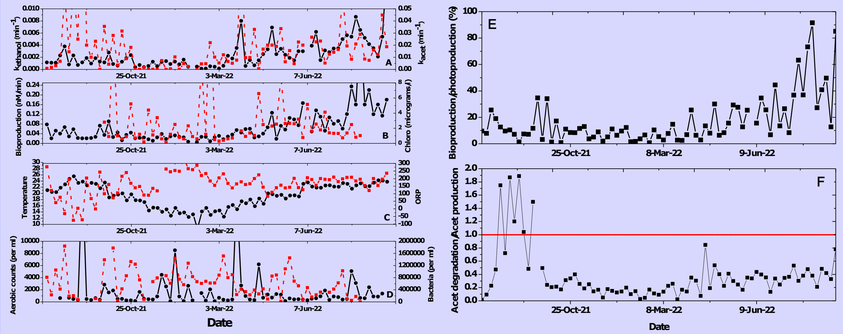


Figure 3. (A-D) Ethanol and acetaldehyde biological degradation rate constants, the rate of dark biological conversion of ethanol to acetaldehyde, some water quality parameters, and bacterial counts as a function of sampling day. (E) Acetaldehyde biological production rate relative to acetaldehyde photochemical production rate. Biological production ranges from 10 to 80% of the photoproduction rate. (F) Acetaldehyde biological degradation relative to photochemical production. Most of the time photoproduction dominates suggesting these waters are a source of acetaldehyde to the atmosphere. Acetaldehyde photochemical production rates were measured and/or calculated from available data in the literature. It was not possible to measure ethanol photochemical production over reasonable times scales in the lab. This suggests that biological degradation dominates and these waters are likely a sink for atmospheric ethanol. The dark biological production of acetaldehyde from ethanol therefore effectively recycles atmospheric ethanol back into the atmosphere as acetaldehyde.

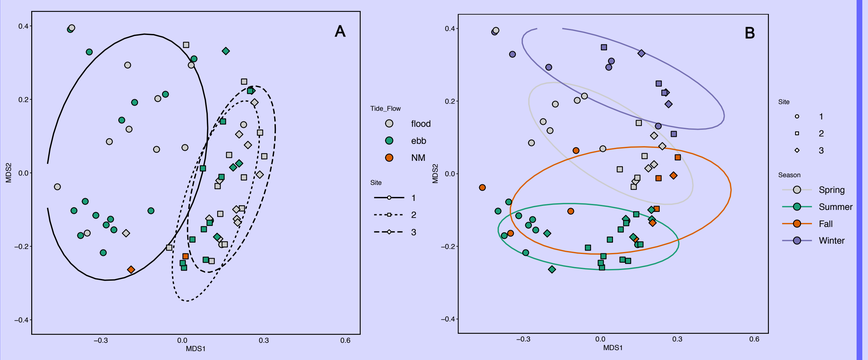


Figure 4. (A) Clustering of microbial composition relative to site. PERMANOVA testing revealed a significant impact of site on composition ($p<0.001$). (B) Clustering of microbial composition relative to season. PERMANOVA testing revealed a significant impact of season on composition ($p<0.001$.

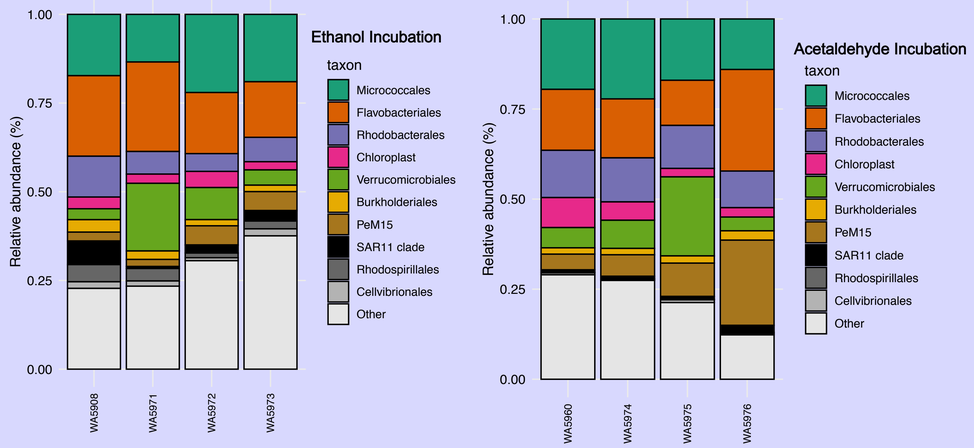


Figure 5. Microbial community composition over several incubation periods. Actinobacterium PeM15 (Acidothermus cellulolyticus) increases and perhaps outcompetes other organisms. Incubations were carried out over the course of a week.

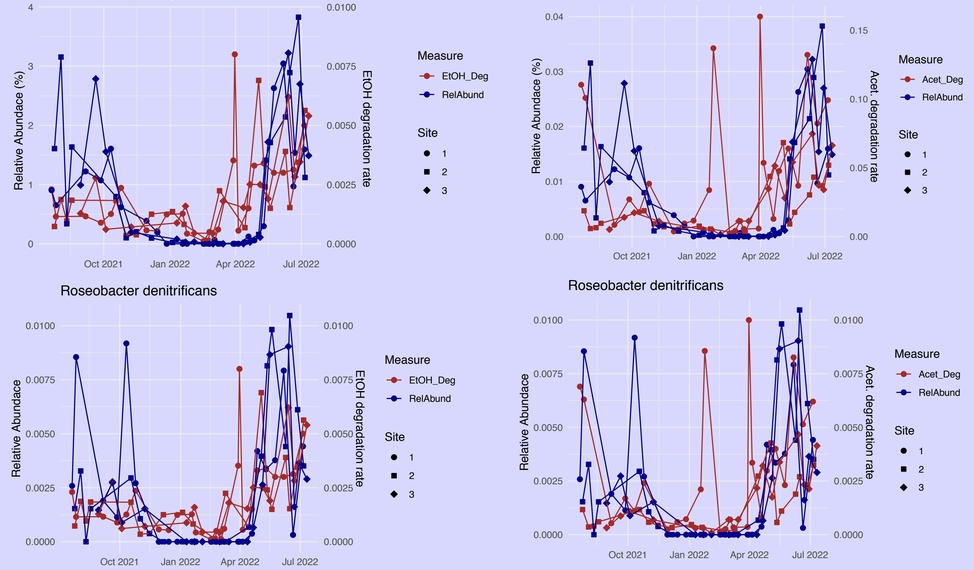


Figure 6. Ethanol and acetaldehyde degredation rates juxtaposed with Acidothermus cellullolyticus and Roseobacter dentrificans abundance.

Acknowledgements

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